

Wiring Beats Blending: Structure-Aware Compensation for Transformer Downscaling

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Abstract

Model families are typically trained size by size, each from scratch. Can a pretrained large model instead be converted into a smaller sibling? We characterize the 1.4B→410M conversion in the Pythia family end to end. Representations align strongly across sizes (ridge $R^2 = 0.84$) while parameters align weakly. Dense weight projection is functionally destructive, and a bit-exact reconstruction control shows this is not an assembly artifact: basis mixing breaks rotary, per-head, GELU, and LayerNorm structure. After the best-fit linear operator, weight residuals carry no learnable or transferable signal under shuffle controls. Conversion value therefore lives in initialization. In matched-budget continued pre-training we decompose conversion into two independent levers: least-squares compensation, a function lever with the best zero-shot quality, and variance-preserving rescale, a dynamics lever with the best endpoints. The decomposition follows a placement rule readable off the architecture: compensation is well-posed exactly on the paths where no normalization sits between the cut and the read (here, the attention-output and MLP down-projections), and the norm-fronted paths take rescale. Compensation is a token-efficient, low-budget win rather than a universal one. At 30M tokens it beats the strongest subcloning variant on both a width-reduced pair (84.0 ± 1.8 vs. 89.7 ± 3.7 , 3/3 seeds) and a held-out depth-reduced pair (109.3 vs. 117.9, 3/3 seeds), reaching a given quality with fewer tokens. At a $33\times$ larger budget the two converge to parity (40.0 vs. 40.0), both far ahead of from-scratch, which transfer initialization always beats: by up to $18\times$ at low budget, with the margin narrowing at convergence and at the largest scale. We also map the method’s boundary. At about $5\times$ the donor scale (6.9B→1.4B) stacking both levers over-corrects, consistent with ill-conditioning of the compensation solve at large width, which points to dimension-aware regularization as a fix. At matched budget our initialization also beats structured pruning with distillation, the standard pipeline for this task, and improves further when combined with it. Code, checkpoints, and the frozen evaluation corpus are released.

1 Introduction

Transformer families ship as ladders of discrete sizes (Pythia alone spans 70M to 12B parameters (Biderman et al., 2023)) because different deployment budgets need different models. In common practice every rung is bought separately: each size is a full pretraining run with its own GPU and energy bill, even though the family members are trained on identical data with a shared tokenizer and architecture, differing only in width and depth. Some recent work instead derives a smaller model from a larger one by structured pruning (Xia et al., 2024; Ma et al., 2023) and knowledge distillation (Hinton et al., 2015), often combined into a single pipeline (Muralidharan et al., 2024); but these keep the donor in the loop: the pruning mask is scored against it and the teacher is queried throughout training. We study the cheaper and less-explored channel in which the donor is used once, at initialization, and then discarded, and we compare against the prune-plus-distill recipe at matched budget (§7). The siblings are, by construction, solutions to the same problem at different capaci-

ties; training each from nothing treats them as unrelated. This paper asks the question the ladder’s cost makes practical and its construction makes tractable: what actually transfers between two sizes of the same pretrained model family, and through what operation can it be carried into the smaller shape?

We answer end-to-end on the Pythia suite, where identical data and tokenizer isolate size as the only variable (§3), in four steps. First, we locate the cross-size relation (§4): the smaller model’s activations are largely a linear image of the larger’s (held-out ridge $R^2 = 0.84$), its weights are not. Second, we diagnose why densely projecting the large weights into the small shape is functionally destructive (§5): a bit-exact control rules out assembly, leaving the projection mathematics, whose basis mixing breaks rotary, per-head, GELU, and LayerNorm structure. Third, what the best-fit linear operator misses carries no learnable signal under shuffle controls (§6), so no zero-shot fix exists at this granularity. Conversion value must therefore live in initialization, and the fourth step is a matched-budget recovery race (§7) that decomposes conversion into two independent levers over structure-respecting selection: closed-form least-squares compensation (repairs what the network computes) and variance-preserving rescale (repairs the scale the optimizer sees), which, acting on disjoint weight paths, stack.

Our contributions are:

- A controlled characterization of what transfers across sizes. Within a family trained on identical data, representations align strongly under a linear map (held-out ridge $R^2 = 0.84$) while parameters align weakly ($R^2 = 0.25$ – 0.39 in the cleanest, embedding case), the gap the rest of the paper studies (§4).
- A mechanically verified diagnosis of why dense projection fails. Rebuilding the donor from its own extracted weights reproduces its logits bit-exactly, so the projected model’s collapse is attributable to basis mixing alone: dense maps break rotary pairing, per-head attention, elementwise GELU, and per-feature LayerNorm, with LayerNorm mismatch the largest single factor in the zero-shot collapse (§5).
- A null result under controls. After the best-fit linear operator, weight residuals carry no cross-layer direction, no patch-learnable signal, and no cross-pair transfer (spectral, consistency, learnability, transfer tests vs. shuffle controls with bootstrap CIs, $p \geq 0.99$): no learned zero-shot correction exists at this granularity (§6).
- A two-lever decomposition of conversion, and a method that stacks them. The levers partition by where normalization sits: least-squares compensation is well-posed only on the paths with no norm between the cut and the read (the attention-output and MLP down-projections), and the norm-fronted paths take variance-preserving rescale — a placement rule readable off any architecture’s wiring. At matched budget, compensation is the function lever (best zero-shot) and rescale the training-dynamics lever (best endpoints); their combination, Compensated Selection, dominates the strongest variant of weight subcloning (Samragh et al., 2023) on both axes and in every seed (84.0 ± 1.8 vs. 89.7 ± 3.7 final perplexity, 3/3 paired wins; zero-shot 18.5k vs. 61.9k) and finishes $18\times$ better than from scratch at 30M tokens; the ordering holds out-of-domain and at $2\times$ the training context, and the margin over from scratch ($\sim 13\times$) generalizes with three seeds to a held-out depth-dominated pair (§7).

2 Related work

Downscaling pretrained models. The closest prior work is Weight Subcloning (Samragh et al., 2023), which initializes a smaller transformer by ranking units of a larger pretrained one, copying selected rows and columns, and rescaling, establishing that selection-based initialization accelerates training of the target size. Sheared LLaMA (Xia et al., 2024) learns structured pruning masks jointly with continued pre-training at a fraction of from-scratch compute; it optimizes the full pipeline, whereas we hold the training recipe fixed across arms and isolate what the initialization alone contributes. Our closed-form compensation belongs to the pruning-with-reconstruction lineage (adjusting surviving weights to absorb

the function of removed ones, from Optimal Brain Surgeon (Hassibi & Stork, 1993) to structured LLM pruning like LLM-Pruner (Ma et al., 2023)), applied once, in closed form, at conversion time. Knowledge distillation (Hinton et al., 2015) transfers function through a teacher’s outputs and requires a full training run; we study the complementary question of what transfers through weights at negligible cost, and the two are combinable.

Growing pretrained models. The mirror direction (initializing a larger model from a smaller one) has a longer history: Net2Net (Chen et al., 2016) introduced function-preserving widening and deepening, bert2BERT (Chen et al., 2022) adapted it to transformers, staged training (Shen et al., 2022) formalized growth operators that preserve loss and dynamics, and LiGO (Wang et al., 2023) learns a structured linear map from small weights to the large initialization. That linear operators carry useful signal upward is consistent with our finding that, downward, even a fitted dense projection initializes far better than random; the asymmetry we document is that dense basis mixing destroys the rotary, per-head, and normalization structure that selection preserves (§5).

Representation similarity and parameter symmetries. Our analyses use standard similarity tools: CKA (Kornblith et al., 2019) and SVCCA (Raghu et al., 2017) measure representation alignment, and model stitching (Lenc & Vedaldi, 2015; Bansal et al., 2021) tests functional interchangeability through a trained adapter. Work on permutation symmetries and model merging (Ainsworth et al., 2023) shows that networks are naturally compared modulo the transformations under which the architecture is invariant. Our selection-versus-blending result is an instance of the same principle: structured selection composes a permutation (an element of the architecture’s symmetry group, restricted to respect head boundaries and rotary frequency pairs (Su et al., 2024)) with coordinate deletion, so every surviving unit keeps the exact nonlinear and positional semantics the architecture assigns it; dense projection mixes coordinates, exits the group, and is punished for it.

Positioning. Each ingredient above exists in isolation: selection-based initialization, least-squares reconstruction, linear growth operators, representation similarity. What is new is the setting (conversion between sizes of a single family trained on identical data, removing data and tokenizer confounds) together with rotary-frequency-matched selection and a controlled end-to-end characterization, from representation alignment through projection diagnosis and residual null tests to a matched-budget race. Because our subcloning baseline re-implements the recipe of Samragh et al. (2023), we disclose three fidelity differences and label it subcloning-style: (a) we score attention heads and MLP units by weight norms and residual lanes by activation variance, where they score by activation magnitudes; (b) when depth must shrink we remove blocks at even stride rather than from the middle; and (c) we evaluate their $\sqrt{d/d'}$ weight rescale directly rather than adopting it wholesale (§7): applied to every matrix it collapses zero-shot quality because LayerNorm already renormalizes most read paths, while its variance-preserving motivation holds exactly on the projections no norm protects, the same two families our compensation re-fits.

3 Experimental setup

Model family and conversion pairs. We study the Pythia suite (Biderman et al., 2023): GPT-NeoX models trained on identical data (the Pile) with a shared tokenizer, isolating size as the only variable between family members. Our primary pair converts 1.4B $\rightarrow$ 410M (24 layers and 16 heads in both; only widths shrink: residual 2048 $\rightarrow$ 1024, head dim 128 $\rightarrow$ 64, MLP 8192 $\rightarrow$ 4096), so layer correspondence is one-to-one and the analysis isolates width. A held-out pair, 410M $\rightarrow$ 160M, exercises the other axes: depth halves (24 $\rightarrow$ 12 blocks), the head count drops (16 $\rightarrow$ 12; head dim unchanged), and width cuts are mild ($\sim$ 25%). A third pair, 6.9B $\rightarrow$ 1.4B, tests scale (all three axes reduced; §7).

Frozen evaluation corpus. All representation analyses probe every model with the same inputs: 10,000 sequences of 128 tokens from the Pile (pile-uncopyrighted), tokenized once and frozen for the project. The first 500 sequences (all token positions) form the similarity

subset (§4); the rest contribute mean-pooled and last-token activations. Continued pre-training streams the same corpus but skips the frozen-set region (contamination guard), and each seed trains on a disjoint stream offset.

Behavioral evaluation. Language-modeling quality is strided sliding-window perplexity on WikiText-103 validation (every token scored once). Hardening evaluations add C4 perplexity (out-of-domain), perplexity at $2\times$ the training context (2048), and zero-shot accuracy on LAMBADA, ARC-Easy, HellaSwag, and PIQA via lm-eval. Reference models anchor every table; anchor scores match published Pythia numbers. All forward passes are greedy and seeded, bit-identical on repetition; weight extraction is validated by exact reconstruction (head-aware QKV split/re-fuse to bit-identical equality; a model rebuilt from its own extracted weights reproduces the original logits with zero difference, §5), so downstream quality changes are attributable to conversion mathematics, not assembly.

Training protocol. Every conversion arm within an experiment shares identical data order, optimizer, cosine schedule, and token budget, so within an experiment the only difference across arms is the starting weights; initialization construction is negligible against any training budget. Full hyperparameters, budgets, and hardware are in Appendix C.

4 Representations align, parameters do not

A size conversion presupposes that the two family members are related; the question is where the relation lives. On the primary width-only pair (1.4B $\rightarrow$ 410M, same depth and head count; §3) we measure the relation in the representations both models compute and in the parameters they store, and find it in only one.

Layer correspondence saturates under CKA. Linear CKA between all 24×24 layer pairs (similarity subset, all token positions; §3) gives a diagonal mean of 0.883, yet only 12.5% of layers place their single best match on the diagonal. The heatmap is one large saturated block (App. Fig. 2, left): layers 0–2 distinct, the middle band (layers ~ 4 –22) mutually ≈ 1.0 , layer 23 distinct, so linear CKA confirms only coarse early/mid/late correspondence (saturation in the middle band is a known limitation of the metric). This motivates a directional map: how much of the small model’s representation is a linear image of the large model’s?

Activations are largely a linear image across widths. Per layer we fit a large $\rightarrow$ small map on pooled activations (8k train / 2k held-out) and score held-out R^2 under two operators: Procrustes (rotation plus one global scale) and ridge (a full linear map $2048 \rightarrow 1024$). Ridge reaches a mean held-out $R^2 = 0.844$ (App. Table 4): the 410M pooled representation is largely a linear projection of the 1.4B’s, capturing 84% of held-out variance. Procrustes already reaches 0.717, so most of the relationship is a change of basis and the ~ 0.13 gap is genuine non-orthogonal reshaping. Both maps dip together at layers 3–5, the layers CKA flagged as distinct (App. Fig. 2, right).

Raw parameters do not. The token embeddings are the one weight matrix where the two sizes share an axis (the vocabulary) and differ only in width, so a single map applies (rows split 80/20). Even in this cleanest case a linear map explains only 0.25–0.39 of weight variance (App. Table 4): ridge $R^2 = 0.248$ on the input embedding and 0.386 on the output embedding, against 0.844 for activations.

The representation–parameter gap. Two independently trained family members reach closely related representations while their parameters are only loosely a linear map of one another. This gap motivates the rest of the paper: if representations align but weights do not, projecting the large weights into the small shape (§5) must leave a substantial residual, and whether that residual is learnable or noise (§6) decides where conversion value comes from.

5 Why dense projection fails

If representations align across sizes, the natural attempt is to project the large model’s weights into the small model’s shape and read off a working small model. We formulate this projection from only the size relationship (never the target’s weights), measure the residual, run the assembled model, and diagnose why it breaks. The verdict: dense blending destroys the network’s structured computation, and a bit-exact control places the destruction in the projection math, not the assembly.

Target-free projection. Each weight is mapped by a pair of structured linear operators,

$$\hat{W} = P_{\text{out}} W P_{\text{in}}^{\top}, \quad P_{\bullet} \in \{P_{\text{res}}, P_{QK}, P_{VO}, P_{\text{MLP}}\}, \quad (1)$$

whose bases are built only from the large model and the size relationship, never from the 410M target. The residual axis uses a global activation-PCA basis P_{res} ; internal axes use SVD bases shared between interacting weights so composed operations stay consistent: Q, K share one, V, O share one, and MLP_UP/MLP_DOWN share a joint basis from the column stack $[\text{UP}, \text{DOWN}^{\top}]$. Sharing is not cosmetic: a per-weight basis for the MLP hidden axis cannot span the target, since one 8192×2048 matrix has rank $\leq 2048 < 4096$, so the interacting pair must be factorized together.

The projected weights miss the target solution. App. Table 5 reports the relative Frobenius error of $\hat{W}$ against the actual 410M weights, mean over 24 layers, for the target-free projection of Eq. 1 and (as a diagnostic upper bound) a shared operator (A, B) fitted with access to the target. Every target-free error exceeds 1 (worse than predicting zero): the projected weights share almost nothing with the 410M’s particular solution, confirming the activation-versus-weight gap of §4 at every layer. Even the fitted operator only reaches 0.66–0.80, so at best $\sim 30\%$ of weight variance is linearly explainable across sizes with one shared map.

Behavioral anchors. We assemble a 410M-shaped model from the projected 1.4B weights and score strided WikiText-103 perplexity against reference and anchor models (App. Table 6). Greedy generation is deterministic but word salad, and the acceptance bar (beat random init) is not met zero-shot. The failure’s shape is instructive: random init produces near-uniform logits (perplexity on the order of vocabulary size), whereas the projected model is confidently wrong, far worse than uniform. This is not a plumbing bug: rebuilding the 1.4B from its own extracted weights with no projection reproduces the real model’s logits with $\max |\text{diff}| = 0$ (bit-exact; previewed in §3), so the entire quality loss in App. Table 6 is attributable to Eq. 1, not to extraction, head-aware QKV split/fuse, embeddings, or LayerNorm plumbing.

Structure mixing breaks the computation. A dense basis change on any internal axis silently changes what the block computes, because every structured or nonlinear operation is tied to specific coordinates. Rotary embeddings act on fixed per-head dimension pairs, so head-space mixing distorts positions. Attention is per-head, yet our global $2048 \rightarrow 1024$ head-space basis mixes dimensions across heads. GELU is elementwise in the MLP hidden space, so $\text{GELU}(Bx) \neq B \text{GELU}(x)$. LayerNorm gains and biases are per-feature, so a rotated stream no longer matches any diagonal gain, which is why the projected-LN variant explodes to 10^{13} .

Separating the violations. The norm violation and the mixing violations can be isolated with arms we already run. Replacing the projected model’s mismatched norms with neutral ones (gain 1, bias 0) recovers $\sim 15\times$ zero-shot ($181,242 \rightarrow 12,323$; App. Table 6, Table 2), so LayerNorm mismatch dominates the immediate collapse.¹ What remains is the basis-mixing cost, and it is a trainability cost more than a zero-shot one: with neutral norms, the projected model starts within 7% of structure-preserving selection (12,323 vs 11,509 at

¹The two numbers come from the strided and quick evaluations respectively; the evals agree at these magnitudes (random init scores $\approx 66\text{k}$ on both).

$t=0$) yet finishes $3\times$ worse after identical training (1,054.5 vs 355.8 at 30M tokens, Table 2). Rotary, per-head, and GELU mixing are not individually isolated here; their joint effect is what the training gap measures. Function-preserving zero-shot conversion therefore needs structure-respecting selection: keep or drop whole heads, hidden units, and features (the architecture’s own symmetry group), not dense blending. Dense projections are useful only as initializations for brief fine-tuning (§7).

6 Post-operator residuals carry no learnable signal

Section 5 shows that even the strongest shared linear operator explains at best $\sim 30\%$ of weight variance (relative error 0.66–0.80), leaving the majority as a residual. Is that residual a structured, learnable correction, or noise? We define the residual per weight as

$$\Delta(l, \text{type}) = W_{410\text{M}} - A W_{1.4\text{B}} B^\top, \quad (2)$$

using the shared fitted operator (A, B) per type, the strongest-alignment condition of §5, so whatever remains is what no single linear size-map can explain and the verdict is conservative: if any signal survives the best linear map, these tests should find it. Four tests probe Δ , each against a matched control.

Test 1: spectral (effective rank). App. Table 7 compares the effective rank of Δ against a per-type shuffled control and a shape/scale-matched Gaussian control. Δ sits 2.8–6.4% below both controls in every type: a faint spectral concentration, so the residual is not perfectly isotropic noise (App. Fig. 3).

Test 2: cross-layer consistency. The mean pairwise cosine of vectorized Δ across layers is $\approx \pm 0.0002$ in every type, identical to the shuffle control: no direction is shared across layers, so no single correction is uniformly missing.

Test 3: learnability (decisive). A patch predictor $R(\hat{W} \text{ patch} + \text{type/layer/position}) \rightarrow \Delta \text{ patch}$ is trained on 18 layers, scored on 6 stratified held-out layers as error reduction over predicting zero; the control retrains on Δ shuffled across layers, with a 1000-sample bootstrap on the real–control gap. Neither a linear nor an MLP-512 predictor beats zero (reductions $-0.0004/-0.0003$), and real and control are indistinguishable (gap $\text{CI}_{95} \approx [-0.0000, -0.0000]$, $p \geq 0.99$; Table 1). No patch-learnable signal.

Test 4: transfer. On the held-out 410M $\rightarrow$ 160M pair the shared operators fit at relative error 0.56–0.74 (smaller width gap). A predictor trained on all primary-pair layers, evaluated on the held-out pair, reduces error by $-0.0004/-0.0003$, again indistinguishable from its shuffle control ($p = 1.00$; Table 1). Nothing transfers, consistent with Test 3.

Reconciling the faint spectral signal. Tests 1 and 3 are not in tension: the 2.8–6.4% spectral deficit is a generic statistical trace, not a predictor-exploitable correspondence, and a behavioral cross-check confirms the destruction lives in the dense projection, not the residual (Appendix B.2).

Handoff to §7. Within a family the sizes share representation geometry but not parameter solutions, and what the best linear size-map misses behaves as noise (no cross-layer direction, no patch-learnable mapping, no cross-pair transfer; only a faint generic spectral fingerprint), so conversion value must come from initialization plus brief fine-tuning, the matched-budget recovery race of §7.

7 Matched-budget conversion: two independent levers

The characterization so far is diagnostic: representations align (§4), dense projection is destructive (§5), and its post-operator residuals carry no learnable signal (§6). The constructive claim follows. If conversion value lives in initialization, the decisive test is a

Table 1: Tests 3–4 (learnability and transfer). Patch-predictor error reduction over predicting zero vs. a label-shuffled control; 1000-sample bootstrap CI on the real–control gap. No predictor beats zero or separates from its control.

Setting	Predictor	Reduction	Shuffle	Gap CI ₉₅	p
Held-out layers (1.4B→410M)	linear	−0.0004	−0.0004	[−0.0000, −0.0000]	0.99
	MLP-512	−0.0003	−0.0003	[−0.0000, −0.0000]	1.00
Transfer (410M→160M)	linear	−0.0004	−0.0004	—	1.00
	MLP-512	−0.0003	−0.0002	—	1.00

CIs are reported for the in-pair held-out test only; the transfer test reports p (M4). Reductions are negative because no predictor improves on predicting zero.

matched-budget recovery race: every candidate gets the same target shape, data, schedule, and token budget, and continued pre-training reveals which starting weights convert fastest. The init decomposes into two independent levers: least-squares compensation, which fixes what the init computes (best zero-shot), and variance-preserving rescale, which fixes the scale the optimizer sees (best endpoints). At a fixed low budget stacking both dominates the strongest single-lever variant on both axes and in every seed, a token-efficiency edge that closes to parity at convergence and inverts at the largest scale (§7.5, §7.6).

Protocol. All arms share the target shape (410M), data order, optimizer, schedule, and token budget; only the starting weights differ (§3). Crucially, every init is obtainable without the trained target: selection scores, compensation moments, and rescale factors are read off the donor 1.4B and the frozen corpus, so no arm leaks target weights. Zero-shot columns are quick-eval perplexity (50k WikiText tokens) at $t=0$; final columns are full strided WikiText-103 perplexity at 30M tokens. Unless a $\pm$ appears, a cell is single-seed (footnoted).

7.1 The ablation ladder

Table 2 walks a seven-arm ladder from an untrained model to the full method. Top to bottom it is monotone in the endpoint (each rung improves the final perplexity), but the zero-shot column is emphatically not monotone, and that dissonance is the section’s main result.

Table 2: Seven-arm ablation ladder, primary pair (1.4B→410M), 30M tokens. Zero-shot is quick-eval perplexity at $t=0$; final is full WikiText-103 perplexity at budget. Rows marked “+” build on the same selection; best zero-shot is hybrid, best endpoint hybrid_rs. All cells single-seed (top two arms re-run across seeds in App. Table 9).

Init	Construction	Zero-shot ppl	Final ppl (30M)
random	from scratch	~66,000	1,519.0
projection	dense linear projection	12,323	1,054.5
subclone	structured selection	11,509	355.8
subclone_iso	+ rescale (read-out paths)	26,669	208.2
hybrid	+ compensation	9,664	114.1
subclone_rs	+ rescale (all cut paths)	61,912	86.4
hybrid_rs	+ compensation + rescale	18,459	83.0

Both transfer inits beat from scratch. Even bare structured selection (subclone, 355.8) recovers $4.3\times$ faster than random (1,519.0) at matched compute, and dense projection (1,054.5), despite its scrambled wiring, still beats from scratch $1.4\times$. Selection preserves valid wiring (rotary pairing, per-head attention, elementwise GELU, per-feature Layer-Norm) and pays off throughout training: subclone separates from projection immediately and the gap widens monotonically (Fig. 1). The full ladder closes at $1,519.0/83.0 = 18\times$ better than from scratch.

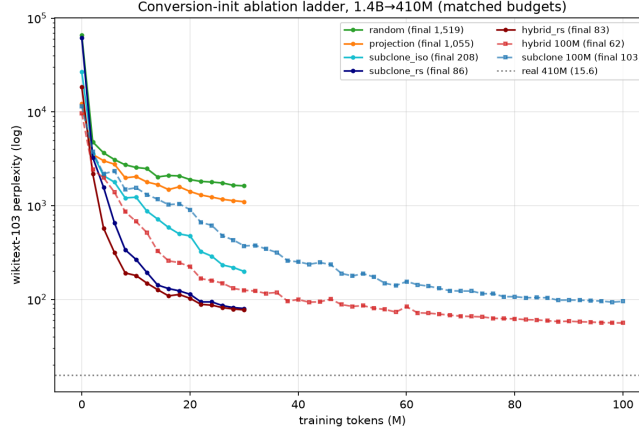


Figure 1: Recovery ladder (1.4B→410M). WikiText-103 perplexity (log scale) vs. tokens, seven arms of Table 2 (dashed = real Pythia-410M, ≈ 15.6 ; 30M tokens = $\sim 0.01\%$ of pretraining, so this measures relative recovery speed).

7.2 Two independent levers

Reading the ladder by construction exposes two orthogonal axes over the selection base.

Compensation is the function lever. The hybrid re-fits the two purely-linear, LayerNorm-free read-out paths (O and the MLP down-projection) by closed-form ridge least squares (§7.4), so surviving units absorb the dropped units’ correlated contribution. This strictly improves both axes: zero-shot $11,509 \rightarrow 9,664$ (best in the table) and endpoint $355.8 \rightarrow 114.1$ ($3.1\times$ over subclone). A better function is better everywhere.

Rescale is the training-dynamics lever. The reference recipe’s variance-preserving factor $\sqrt{d_{\text{in}}/d'_{\text{in}}}$ (Samragh et al., 2023) does the opposite. On the LayerNorm-free read-out paths only (subclone_iso) it worsens zero-shot $11,509 \rightarrow 26,669$ yet improves the endpoint $355.8 \rightarrow 208.2$; on every input-cut path (subclone_rs) it collapses zero-shot to 61,912 ($5.4\times$ worse than bare selection) while driving the endpoint to 86.4, past even the compensated hybrid. It does not fix the function (a downstream LayerNorm renormalizes each rescaled read path, so the scalar cancels in the forward pass and merely perturbs zero-shot alignment), but it fixes the variance the optimizer starts from, and along this axis the trade is monotone: more rescale, worse start, better endpoint.

The levers stack. The two levers act on disjoint quantities: compensation on the function of the two LN-free read-out paths, rescale on the scale of the LN-fronted read-in paths (Q, K, V , MLP up-projection, output embedding), so they compose. hybrid_rs applies compensation where it is ridge-optimal and the reference-recipe scale where compensation cannot reach, dominating the strongest single-lever arm (subclone_rs) on both axes: zero-shot $18,459 < 61,912$ and endpoint $83.0 < 86.4$.

7.3 Seeded verdict

The two rescale leaders are close on the single seed (86.4 vs. 83.0), so we re-run both across three data-draw seeds. The separation is clean: hybrid_rs (84.0 ± 1.8 final WikiText-103 perplexity) wins every paired comparison against subclone_rs (89.7 ± 3.7) (3/3), and its worst seed (86.0) still beats subclone_rs’s best seed (86.4), so the seed distributions do not overlap (per-seed detail in App. Table 9).

This upset (subclone_rs at 86.4 beating the compensation-only hybrid at 114.1) corrects our own earlier reading that compensation subsumes rescale; the reference recipe is vindicated

exactly on the LN-fronted read-in paths compensation leaves alone, so the principled method is the stack, not either lever (Appendix B.1).

7.4 Compensated Selection

Algorithm 1 gives the full construction (per-coordinate scoring in the listing). Selection keeps whole coordinates so each subspace’s symmetry survives: with rotary half-width $16 \rightarrow 8$ the small frequency ladder is exactly every other large frequency, so we keep rotary pairs at stride 2 to preserve the (cos, sin) pairing (Su et al., 2024). When the head count shrinks (the held-out pair) selection is promoted to whole heads shared by Q, K, V, O ; when depth shrinks, evenly strided blocks are kept with their norms and biases.

Compensation is the closed-form optimum for a pruned linear map. For a map M reading a signal with measured second moment Σ , the ridge least-squares substitute on the kept columns $\mathcal{K}$ is

$$M' = \arg \min_{M'} \mathbb{E} \|Mx - M'x_{\mathcal{K}}\|^2 = M \Sigma_{:, \mathcal{K}} (\Sigma_{\mathcal{K}, \mathcal{K}} + \lambda I)^{-1}, \quad (3)$$

with plain subcloning as the special case $M' = M_{:, \mathcal{K}}$. We apply it at the two spots where the reconstruction target is well-defined (O and the MLP down-projection, both purely linear with no normalization between cut and read), and leave the LN-fronted read-in paths to the optional rescale lever. Least-squares reconstruction after pruning is classical (Hassibi & Stork, 1993; Ma et al., 2023); new here are the setting (family size conversion), the rotary-frequency-matched selection, and the two-lever decomposition.

Algorithm 1 Compensated Selection (hybrid_rs)

Require: donor weights W_L ; frozen-corpus activation variances and second moments Σ ; target shape (residual d' , head dim, MLP width, block map)
 Ensure: installable small-model weights (no target-model access)
 1: $\mathcal{R} \leftarrow$ top- d' residual features by mean-over-layer activation variance {one global basis}
 2: for each kept block b do
 3: $\mathcal{K}_{qk} \leftarrow$ frequency-matched rotary pairs (stride 2) $\cup$ top non-rotary dims by $\|q\| + \|k\|$
 4: $\mathcal{K}_{vo} \leftarrow$ top head dims by $\|V_{\text{row}}\| \cdot \|O_{\text{col}}\|$ {whole heads if head count shrinks}
 5: $\mathcal{K}_{mlp} \leftarrow$ top units by $\|\text{up}_{\text{row}}\| \cdot \|\text{down}_{\text{col}}\|$
 6: slice W_L block b to $(\mathcal{R}, \mathcal{K}_{qk}, \mathcal{K}_{vo}, \mathcal{K}_{mlp})$; LayerNorms and biases ride along
 7: $O' \leftarrow \text{LScomp}(O, \Sigma_{\text{attn-in}}, \mathcal{K}_{vo})$; $\text{MLP_down}' \leftarrow \text{LScomp}(\text{MLP_down}, \Sigma_{\text{mlp-in}}, \mathcal{K}_{mlp})$
 {Eq. 3, function lever}
 8: if rescale then
 9: multiply LN-fronted read-in paths $Q, K, V, \text{MLP_up}$ (and output embedding) by $\sqrt{d_{\text{in}}/d'_{\text{in}}}$
 {dynamics lever}
 10: end if
 11: end for
 12: return assembled small model

7.5 Hardening

The headline is one metric at one budget on one pair; we stress it on all three.

Every metric, same ordering. App. Table 8 adds C4 perplexity (out-of-domain), perplexity at $2\times$ the training context (2048), and four zero-shot tasks. The ordering hybrid $<$ subclone $<$ projection $<$ random holds on every metric. Perplexity at 2048 tracks each model’s 1024 number, so rotary-frequency-matched selection extrapolates past its training length without penalty. LAMBADA (floor ≈ 0) is the sharpest ability gap (hybrid $14\times$ subclone); the multiple-choice tasks sit near guess floors, so 30M tokens buys language modeling, not task reasoning; anchor scores match published Pythia numbers.

The lead persists at longer budget. Re-running the two leaders at 100M tokens, hybrid finishes at 62.3 and subclone at 103.1; hybrid crosses subclone’s 100M endpoint at ~ 38 M tokens, a $\sim 2.6\times$ token-efficiency gain.²

At convergence, the edge closes. Pushing the primary pair to 1B tokens ($33\times$ the headline budget), the two rescale leaders finish level: hybrid_rs 40.0 vs. subclone_rs 40.0 (parity in this run; one seed per arm), both $\sim 1.4\times$ ahead of from scratch (57.4) and far nearer the real 410M (≈ 15.6) than at 30M. Compensation’s advantage is thus a low-budget, token-efficiency effect: it reaches a given quality with fewer tokens but is redundant with rescale once the budget is large, the two levers improving the same underlying conditioning so that at convergence one suffices.

A held-out pair, and the method’s boundary. The 410M $\rightarrow$ 160M pair exercises axes the primary pair does not (depth-dominated: depth halves, head count $16 \rightarrow 12$ with whole-head selection, mild width cuts; §3). Across three seeds (final WikiText-103 perplexity at 30M tokens: hybrid 113.2 ± 4.5 , subclone 118.5 ± 3.3 , random $1,505.2 \pm 57.9$; per-seed detail in App. Table 10), transfer beats from scratch $\sim 13\times$ on every seed, so the core claim generalizes, with error bars, to an untuned regime. The full stack keeps its low-budget edge here too: hybrid_rs (109.3) beats subclone_rs (117.9) on all three seeds. Compensation alone, though, is neutral: hybrid and subclone here tie (overlapping $\pm 1\sigma$), unlike the $3.1\times$ compensation gap on the width pair. Compensation’s width-repair edge is neutral when damage is depth-dominated (§9), costing nothing here (zero-shot starts at hybrid ≈ 30 k vs. 9.7k on the width pair). The tie is budget-robust (100M tokens, seed 0: hybrid 77.1 vs. subclone 74.4, a 3.6% gap inside seed noise, leader flipped) and holds on every other metric, with random far below and the real-160M anchor matching published Pythia numbers (single-seed; App. A.6).

7.6 Scale, and the method’s boundary

The scale pair (6.9B $\rightarrow$ 1.4B) reduces all three axes at once at $\sim 5\times$ the donor size (30M tokens; App. Table 11). Here the stack inverts: rescale-only subclone_rs is best (572), compensation-only hybrid next (776), and stacking both (hybrid_rs, 1,213) is worst among transfer arms, though all still beat from scratch (1,413), so the core claim survives its worst case. Each lever alone is safe; only their combination over-corrects, and only at this scale. A preliminary control, converting a smaller (1.4B) donor to targets at the same reduction ratios, did not reproduce the over-correction, suggesting the cause is absolute scale rather than the reduction geometry: the compensation solve (Eq. 3) inverts a far larger, worse-conditioned matrix at 8192-wide inputs than in the smaller cases (≤ 4096), which its magnitude-scaled ridge under-regularizes. A controlled study across donor scales, and the dimension-aware fix, are left to future work (§9).

Compute accounting. Initialization construction is negligible and included in the accounting (selection: norms plus top- k , seconds; compensation moments over 1,000 frozen-corpus sequences, ~ 75 s; closed-form solves, seconds; rescale, free). Every arm then trains on the same 30M tokens under the same schedule (~ 55 min each on a single GB10 at ~ 9.4 k tok/s), so all standings (Table 2; App. Table 9) compare equal-budget endpoints differing only in starting weights.

7.7 Comparison to knowledge distillation

Structured pruning followed by knowledge distillation is the standard pipeline for building a small model from a large one (Muralidharan et al., 2024; Xia et al., 2024). Our selection step is itself a structured pruning (§7.2), so subclone_rs trained with a distillation loss is that recipe reproduced in our controlled setting; the question is whether the two-lever init still helps once a teacher is in the loop. We add a distillation objective to continued pre-training, $L = \frac{1}{2} \text{CE} + \frac{1}{2} T^2 \text{KL}(\text{student}||\text{teacher})$, with the donor 1.4B as a frozen teacher

²100M cells are single-seed; cross-horizon comparisons are schedule-confounded (finals compared only within a horizon).

and $T=2$, and re-run the primary pair at 30M tokens. The teacher forward caps the batch at 16, so we run the no-distill arms at batch 16 as well for a matched comparison (Table 3); the batch-32 finals of Table 2 agree on the ordering.

Table 3: Adding knowledge distillation (frozen 1.4B teacher) to continued pre-training, primary pair (1.4B→410M), 30M tokens, batch 16 throughout for a matched comparison. Final full WikiText-103 perplexity, mean±std over three data-draw seeds (random: single seed); subclone_rs is a structured-pruning init, so its distilled column is the prune-plus-distill recipe.

Init	no distillation	+ distillation
random	1,359.4	1,311.3
subclone_rs (pruning)	93.9 ± 2.0	103.5 ± 7.4
hybrid_rs (ours)	88.2 ± 1.2	80.8 ± 2.1

Three findings, each holding in all three seeds. First, the two-lever init beats the pruning init with and without a teacher (hybrid_rs 88.2 ± 1.2 vs subclone_rs 93.9 ± 2.0 undistilled; 80.8 ± 2.1 vs 103.5 ± 7.4 distilled; 3/3 paired wins in both columns), so the contribution is not subsumed by distillation. Second, the two are complementary: distillation improves hybrid_rs in every seed ($88.2 \rightarrow 80.8$ on average) but degrades the pruning init in every seed ($93.9 \rightarrow 103.5$), which over-corrects under the added objective much as the stacked levers do at the largest scale (§7.6), and hybrid_rs+distillation is the best arm in every seed. Third, distillation is not free: querying the teacher every step halves throughput ($\sim 4.5k$ vs $\sim 9.4k$ tok/s), whereas our init reads the donor once at construction and never again. We do not reproduce the billion-token, multi-billion-parameter results of Muralidharan et al. (2024); Xia et al. (2024); this is a matched-budget comparison in one controlled setting that isolates the initialization’s contribution.

8 Conclusion

We asked what actually transfers when a pretrained transformer is converted into a smaller sibling, and answered it in a controlled, single-family setting: representations align across sizes but parameters do not, dense weight projection is destructive for a mechanically verified reason: it breaks the architecture’s essential wiring, and after the best-fit linear operator the weight residuals carry no signal our probes can exploit. The value of a conversion therefore lives in initialization, not in any learnable zero-shot weight correction. Decomposing that initialization into two independent levers, least-squares compensation (what the network computes) and variance-preserving rescale (the scale the optimizer sees), lets us state the practical rule precisely: compensation is a token-efficient, low-budget win that stacks with rescale to dominate subcloning at small budgets on both a width- and a depth-reduced pair, ties it once the budget is large enough to converge, and, stacked, over-corrects only at the largest donor scale, where the compensation solve is ill-conditioned. For a practitioner spinning up a new size, the takeaway is concrete: initialize by structure-respecting selection, compensate and rescale on disjoint paths for a strong low-budget start, and fall back to the single robust rescale lever at large scale until a dimension-aware regularizer closes that gap.

9 Limitations and future work

One family, one wiring. All experiments are within the Pythia suite (GPT-NeoX: Layer-Norm, elementwise GELU, rotary attention), a deliberate choice (identical data and tokenizer isolate size), but the two-lever decomposition leans on those wiring details: which paths a norm fronts sets where each lever is well-posed (§7.4). Llama-/Qwen-class models change the essential parts (RMSNorm, gated SwiGLU, grouped-query attention), so porting needs a per-architecture safe-wiring analysis; the characterization tools are architecture-agnostic.

Budgets, seeds, and open directions. Our recovery budgets (30M primary, 100M persistence, 1B convergence) span $\sim 0.01\text{--}0.3\%$ of Pythia’s 300B-token pretraining; at 1B tokens converted models reach 40.0 (vs. the real 410M’s ≈ 15.6), much closer than at 30M (83.0) but still short of full pretraining, so most claims concern relative recovery at matched budget rather than end quality. Decisive comparisons carry three data-draw seeds (App. Tables 9 and 10); remaining cells are single-seed. Two directions stay open. First, compensation touches only the two LN-free read-out paths; on the LN-fronted read-in paths the same renormalization that collapses blanket-rescale zero-shot (§7.2) makes the least-squares target ill-defined, so normalization-aware compensation there is open. Second, the held-out pair marks a harder boundary (§7.5): compensation repairs width cuts but not deleted blocks; reduction-aware conversion and width-vs-depth budget allocation (our boundary favors width) are untried.

Compensation conditioning at scale. The two-lever stack, which dominates at low budget on both smaller pairs, over-corrects on the 6.9B $\rightarrow$ 1.4B pair (§7.6). The least-squares compensation (Eq. 3) inverts a second-moment matrix whose conditioning worsens with the kept width, and its magnitude-scaled ridge under-regularizes the low-variance directions at 8192-wide inputs. A preliminary small-donor control at matched reduction ratios did not reproduce the effect, suggesting the cause is absolute scale rather than the reduction geometry. Two remedies are open for large donors: a spectrum- or dimension-aware ridge (setting λ from the eigenvalue spread rather than the mean diagonal, or clamping the compensated-weight norm), and a stronger, importance-based selection criterion (gradient- or Hessian-scored, as in structured pruning) in place of magnitude selection. Both are untested here; until then, the robust single-lever `subclone_rs` is the safe default at large donor scale.

Reproducibility statement

All experiments use the public Pythia suite and the pile-uncopyrighted, WikiText-103, and C4 corpora. Every result comes from a seeded, deterministic pipeline: weight extraction is validated by exact (bit-identical) reconstruction, forward passes are greedy and reproducible per device, and each reported comparison fixes data order, optimizer, schedule, and token budget across arms — varying only the initialization. The frozen $10,000 \times 128$ evaluation corpus is built once from a recorded stream offset and contamination-guarded against the continued-pre-training stream; decisive comparisons report three data-draw seeds. The full code, all configs and seeds, the `uv.lock` environment pin, the frozen-corpus recipe, and the released checkpoints will be made publicly available at <https://github.com/rsdpynugula/ScaleOp> upon publication.

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A Deferred tables and figures

This appendix collects tables and figures deferred from the main text for space. All numbers are referenced from, and discussed in, the sections indicated.

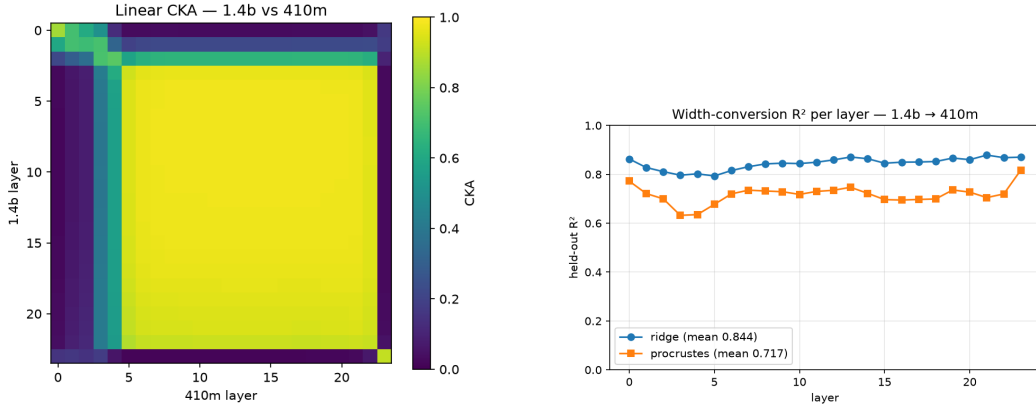


Figure 2: Alignment across sizes for the primary pair (1.4B $\rightarrow$ 410M). (left) Linear CKA between all 24×24 layer pairs: a saturated middle band (layers ~ 4 –22 mutually ≈ 1.0) confirms coarse correspondence but resolves no sharp per-layer match. (right) Held-out R^2 of the per-layer activation maps: ridge (full linear) and Procrustes (rotation+scale) track each other and dip together at layers 3–5, the layers CKA also flags as distinct.

Table 4: Alignment across sizes, held-out. Activations align strongly under a linear map; raw parameters, even in the cleanest (embedding) case, do not. Best (highest) map per row in bold.

Target	CKA	Ridge R^2	Procrustes R^2
Pooled activations (per-layer mean)	0.883 [†]	0.844	0.717
Input embedding	0.512	0.248	0.167
Output embedding (unembedding)	0.678	0.386	0.242

[†] For activations, the diagonal mean of the 24×24 linear-CKA matrix, which saturates in the middle band (Figure 2, left); for embeddings, linear CKA before mapping.

A.1 Alignment across sizes (§4)

A.2 Projection diagnostics (§5)

A.3 Spectral residual test (§6)

A.4 Extended evaluation, primary pair (§7.5)

A.5 Primary-pair seeded verdict (§7.3)

A.6 Held-out pair per-seed detail (§7.5)

Metric robustness (held-out pair). On the held-out 410M $\rightarrow$ 160M pair the hybrid-vs-subclone tie holds on every metric beyond WikiText-103: C4 perplexity 91.1 vs. 94.2, perplexity at $2\times$ context 111.8 vs. 110.2, LAMBADA 11.0% vs. 10.7% (others within noise), while random sits at C4 514 and LAMBADA 0; the real-160M anchor (LAMBADA 35.4%) matches published Pythia numbers. These cells are single-seed.

Table 5: Relative Frobenius error of projected weights vs. the actual 410M weights (mean over 24 layers). Every target-free error exceeds 1 (worse than zero); lower per row in bold.

Weight	Projection (target-free)	Operator [†] (fitted)
Q	1.471	0.658
K	1.591	0.758
V	1.390	0.804
O	1.724	0.795
MLP_UP	1.668	0.683
MLP_DOWN	1.723	0.686

[†] A single (A, B) shared across layers per type; it peeks at the target and so upper-bounds the linearly explainable fraction. A per-layer fit is degenerate (error $\rightarrow 0$ around any full-rank W) and is not reported.

Table 6: Zero-shot dense-projected quality (strided WikiText-103 perplexity); the projected model does not beat random init. Best (lowest) in bold.

Model	Perplexity
real 1.4B (reference)	11.5
real 410M (upper anchor)	15.6
random-init 410M (lower)	65,962
proj., LN from 410M	181,242
proj., LN from 1.4B	10^{13}
proj., neutral LN*	12,323
structured selection*	11,509

*Quick-eval perplexity (the two evals agree at these magnitudes: random init scores $\approx 66k$ on both). Neutral norms (gain 1, bias 0) recover $\sim 15\times$, so LayerNorm mismatch dominates the zero-shot collapse; the residual gap to structured selection is small at $t=0$ but $3\times$ after training — the decomposition in §5.

Table 7: Test 1 (spectral). Effective rank of the residual Δ vs. a per-type shuffled control and a shape/scale-matched Gaussian control. Δ sits 2.8–6.4% below both in every type — a faint, non-isotropic concentration, not a learnable direction.

Weight	Δ erank	Shuffled	Gaussian
Q	771.3	824.1	824.1
K	784.0	824.0	824.1
V	801.2	824.2	824.1
O	792.2	824.1	824.1
MLP_UP	947.2	989.4	989.4
MLP_DOWN	934.8	989.4	989.4

A.7 Scale pair (§7.6)

B Deferred discussion

B.1 An honest correction (§7.3)

The rescale lever corrects our own earlier reading. When the hybrid first won at 114.1 we argued that least-squares compensation subsumed the reference recipe’s scalar rescale, since a ridge-optimal map beats a scalar on the paths it re-fits. The upset — subclone_rs at 86.4, beating the hybrid with no compensation — showed that is only half right: compensation dominates rescale on the two read-out paths it touches, but the LN-fronted read-in paths it leaves alone still benefit from the reference recipe’s variance correction as a training-dynamics prior. Blanket rescale collapses zero-shot yet wins on dynamics, so the reference recipe is vindicated exactly where our first account wrote it off, and the principled method is the stack, not either lever alone.

B.2 Reconciling the faint spectral signal (§6)

Tests 1 and 3 are not in tension: the 2.8–6.4% spectral deficit is a generic statistical trace (mild row/column-norm heterogeneity), not a correspondence between $\hat{W}$ and Δ any predictor can exploit. A behavioral cross-check agrees: an assembled 410M whose six matrix types are operator-projected (with real biases, LayerNorms, and embeddings) scores 35,212

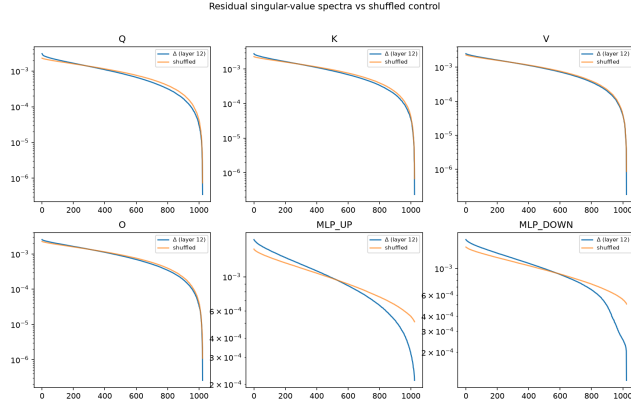


Figure 3: Test 1 (spectral). Singular-value spectrum of the residual Δ against shuffled and Gaussian controls. Δ concentrates 2.8–6.4% more than either control in every weight type — a faint, generic non-isotropy that carries no layer-specific, predictable signal (Table 1).

Table 8: Extended evaluation of the 30M primary-pair checkpoints and the real-410M reference. Guess floors: arc_easy 0.25, hellaswag 0.25, piqa 0.50, lambada ≈ 0 (open-vocabulary). Best converted arm in bold; the ordering holds on all six metrics. Single-seed checkpoints. This table predates the rescale-lever ablation and reports the compensation contrast (hybrid vs. subclone); the rescale gains of Tables 2–9 stack on top of it.

Model	C4 ppl	wk@2048	lambada	arc_easy	hellaswag	piqa
hybrid	90.7	109.8	0.076	0.313	0.266	0.555
subclone	162.7	346.3	0.005	0.308	0.263	0.548
projection	380.5	1,210	0.000	0.272	0.258	0.539
random	518.8	1,788	0.000	0.264	0.256	0.532
real 410M	21.0	14.5	0.516	0.519	0.337	0.667

perplexity, and the learned correction moves it only $\sim 2\%$ (to 34,373); both remain non-functional. Notably the operator matrices alone beat random init (35k vs. 66k) once the surrounding tensors are real — unlike §5’s fully projected model — again locating the destruction in the dense projection, not the residual. (The correction’s gains on the 18 training layers are partly memorized; only the held-out layers carry genuine predictions, and there the reduction is zero.)

C Training protocol details

Every conversion arm within an experiment shares identical data order, optimizer (AdamW, $\beta = (0.9, 0.95)$, weight decay 0.1), cosine schedule with 100-step warmup, gradient clipping at 1.0, bf16 autocast, and token budget (30M primary; 100M persistence checks); the learning rate equals the target size’s original pretraining rate. Initialization construction costs are negligible against any training budget (selection: seconds; moments pass for compensation: ~ 75 s; closed-form solves: seconds) and are included in the compute accounting. All experiments ran on a single NVIDIA GB10 (128 GB unified memory); any 24 GB+ CUDA GPU reproduces them.

Table 9: Primary-pair final WikiText-103 perplexity across three data-draw seeds (30M tokens; identical budget/schedule per seed). hybrid_rs wins every paired comparison; its worst seed beats subclone_rs’s best.

Init	seed 0	seed 1	seed 2	mean $\pm$ std
subclone_rs	86.4	93.7	88.9	89.7 ± 3.7
hybrid_rs	83.0	86.0	82.8	84.0 ± 1.8

Table 10: Held-out pair (410M→160M), final WikiText-103 perplexity across three data-draw seeds (30M tokens). Transfer beats from scratch $\sim 13\times$ on every seed; hybrid and subclone are a statistical tie on this depth-dominated pair (overlapping $\pm 1\sigma$).

Init	seed 0	seed 1	seed 2	mean $\pm$ std
hybrid	117.5	113.6	108.5	113.2 ± 4.5
subclone	115.0	121.6	118.9	118.5 ± 3.3
random	1,464.4	1,571.5	1,479.6	$1,505.2 \pm 57.9$

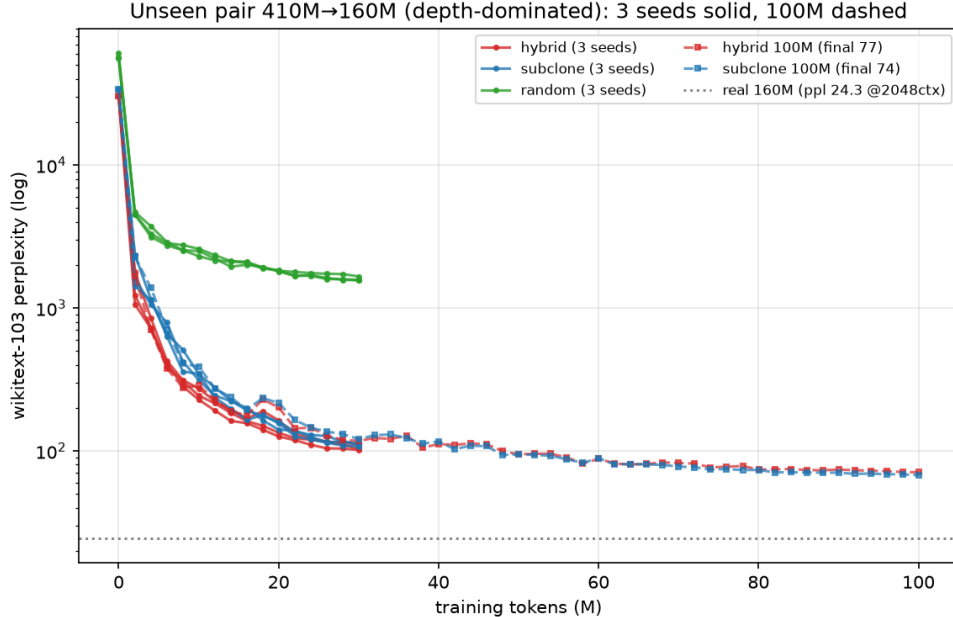


Figure 4: Held-out pair (410M→160M), three seeds. Full WikiText-103 perplexity (log scale) versus tokens; bands span the three data-draw seeds and the dashed line marks the real Pythia-160M reference. Transfer inits (hybrid, subclone) sit an order of magnitude below random throughout and overlap each other — the depth-dominated regime where compensation’s width-repair edge is neutral (Table 10), yet the $\sim 13\times$ margin over from scratch is preserved on every seed.

Table 11: Scale pair (6.9B→1.4B), 30M tokens, single seed. The two levers that stack at smaller scale here anti-synergize — each alone beats the combination — though every transfer arm still beats from scratch (§7.6).

Init	Construction	Final ppl (30M)
random	from scratch	1,413
hybrid_rs	compensation + rescale	1,213
hybrid	compensation only	776
subclone_rs	rescale only	572